

# Arithmetic in Spark `shockDetect`

Source: `source/physics/Hydro/HydroMain/Spark/Hydro_funcs.F90`, routine `shockDetect` (lines 436–570). The routine flags cells with both a sufficiently large pressure variation and sufficiently compressive velocity differences.

## 1 Notation, domain, and initialization

Let  $q = (i, j, k)$  denote a cell,  $N = \text{NDIM}$ , and  $\mathbf{e}_d$  a unit index offset in direction  $d$ . Let  $p_q = U(\text{PRES}, q)$ ,  $\rho_q = U(\text{DENS}, q)$ ,  $\gamma_q = U(\text{GAMC}, q)$ , and  $v_{d,q}$  be the stored velocity component. The array  $U$  denotes `solnData`. Let  $\mathcal{G}$  be the full guard-cell-inclusive region and  $\mathcal{L}$  the region supplied in `limits`.

The routine returns immediately if `SHOK_VAR` is not defined. Otherwise, it first clears the shock flag over the entire pointed-to array:

$$S_q \leftarrow 0 \quad \text{for all cells in } \text{solnData}. \quad (1)$$

The two fixed detection parameters in this implementation are

$$\beta = 0.5, \quad \delta = 0.1. \quad (2)$$

## 2 Sound speed and neighborhood minima

The temporary array `Vc` holds the sound speed at every cell in  $\mathcal{G}$ :

$$c_q = \sqrt{\frac{\gamma_q p_q}{\max(\rho_q, \tau)}}, \quad \tau = \text{hy\_tiny}. \quad (3)$$

Only the density denominator is floored; the numerator and square-root argument receive no additional clamp.

For each  $q \in \mathcal{L}$ , define the neighborhood

$$\mathcal{N}_N(q) = \left\{ q + \sum_{d=1}^N s_d \mathbf{e}_d : s_d \in \{-1, 0, 1\} \right\}. \quad (4)$$

This contains 3, 9, or 27 cells for one, two, or three dimensions, respectively, including the central cell and diagonal neighbors. The threshold reference values are

$$p_{\min,q} = \min_{q' \in \mathcal{N}_N(q)} p_{q'}, \quad c_{\min,q} = \min_{q' \in \mathcal{N}_N(q)} c_{q'}. \quad (5)$$

The supplied limits must permit these one-cell neighbor accesses within the available data and temporary sound-speed array.

### 3 Undivided velocity and pressure differences

The quantity stored in `divv` is

$$D_q = \frac{1}{2} \sum_{d=1}^N (v_{d,q+e_d} - v_{d,q-e_d}). \quad (6)$$

For example, in three dimensions,

$$D_{i,j,k} = \frac{1}{2} [v_{x,i+1,j,k} - v_{x,i-1,j,k} + v_{y,i,j+1,k} - v_{y,i,j-1,k} + v_{z,i,j,k+1} - v_{z,i,j,k-1}]. \quad (7)$$

The directional pressure differences are

$$g_{d,q} = \begin{cases} \frac{1}{2} (p_{q+e_d} - p_{q-e_d}), & d \leq N, \\ 0, & d > N, \end{cases} \quad d \in \{x, y, z\}. \quad (8)$$

Their combined magnitude is the sum of absolute directional differences:

$$G_q = |g_{x,q}| + |g_{y,q}| + |g_{z,q}|. \quad (9)$$

These are undivided differences: neither  $D_q$  nor  $g_{d,q}$  is divided by the cell spacing. No cylindrical or spherical geometry factors are used. Thus  $D_q$  has velocity units, and  $G_q$  has pressure units.

### 4 Detection switches and output flag

The two switches are initialized to false for each evaluated cell, then set according to the inclusive comparisons

$$\text{SW1}_q = [G_q \geq \beta p_{\min,q}], \quad (10)$$

$$\text{SW2}_q = [D_q \leq -\delta c_{\min,q}]. \quad (11)$$

The final shock flag is

$$S_q = \begin{cases} 1, & q \in \mathcal{L}, \quad G_q \geq 0.5 p_{\min,q} \text{ and } D_q \leq -0.1 c_{\min,q}, \\ 0, & \text{otherwise.} \end{cases} \quad (12)$$

For positive reference pressure and sound speed, the first condition requires strong pressure variation and the second requires compression. Reducing either threshold parameter makes its condition easier to meet. The routine writes only the flag; changes to reconstruction or Riemann solver selection occur elsewhere.